

Quiz 3: Conditional Expectation, Entropy, Stochastic Processes. Solutions

Homework for lectures 5 and 6. Notation as in the lectures. Twelve problems in three parts, four per part, numbered 1–12.

Part A: Conditional Expectation

Notation as in the Conditional Expectation lecture. For random variables X, Y on $(\Omega, \mathcal{F}, \mathbb{P})$, $\mathbb{E}[X \mid Y = y]$ is the average of X over the row $\{Y = y\}$, and $\mathbb{E}[X \mid Y]$ is the random variable that takes the value $\mathbb{E}[X \mid Y = y]$ on the event $\{Y = y\}$. The results used below are the law of total expectation, the tower property, “taking out what is known”, and the restart property.

1. A random defect rate. A supplier’s defect rate is unknown. Model it as a random variable Θ with the uniform distribution on $[0, \frac{1}{5}]$; its pdf is $f_\Theta(\theta) = 5$ for $0 \leq \theta \leq \frac{1}{5}$. A sample of 10 items is inspected. Let D be the number of defective items in the sample. Given $\Theta = \theta$, D is binomial with parameters 10 and θ . Take the following three facts about the binomial distribution as given:

$$\begin{aligned}\mathbb{E}[D \mid \Theta = \theta] &= 10\theta, \\ \mathbb{E}[D^2 \mid \Theta = \theta] &= 10\theta(1 - \theta) + 100\theta^2, \\ \mathbb{P}(D = 0 \mid \Theta = \theta) &= (1 - \theta)^{10}.\end{aligned}$$

- (a) Compute $\mathbb{E}[D]$.
- (b) Compute $\mathbb{E}[D^2]$ and $\text{Var}(D)$.
- (c) A binomial random variable with parameters 10 and 0.1 has variance $10 \cdot 0.1 \cdot 0.9 = 0.9$. State whether $\text{Var}(D)$ is larger or smaller than 0.9. Give the reason in one sentence.
- (d) Compute $\mathbb{P}(D = 0)$ by integrating $\mathbb{P}(D = 0 \mid \Theta = \theta)$ against the pdf of Θ . Compare the result with 0.9^{10} .

Solution. (a) By the law of total expectation, $\mathbb{E}[D] = \int_0^{1/5} \mathbb{E}[D \mid \Theta = \theta] f_\Theta(\theta) d\theta = 10 \mathbb{E}[\Theta]$. The mean of the uniform distribution on $[0, \frac{1}{5}]$ is the midpoint, $\mathbb{E}[\Theta] = \frac{1}{10}$.

$$\mathbb{E}[D] = 1.$$

(b) First $\mathbb{E}[\Theta^2] = \int_0^{1/5} 5\theta^2 d\theta = \frac{5}{3} \cdot \frac{1}{125} = \frac{1}{75}$. By total expectation, $\mathbb{E}[D^2] = \mathbb{E}[10\Theta(1 - \Theta) + 100\Theta^2] = 10 \mathbb{E}[\Theta] + 90 \mathbb{E}[\Theta^2] = 1 + \frac{90}{75} = \frac{11}{5}$. Then $\text{Var}(D) = \mathbb{E}[D^2] - \mathbb{E}[D]^2 = \frac{11}{5} - 1 = \frac{6}{5}$.

$$\mathbb{E}[D^2] = 2.2 \text{ and } \text{Var}(D) = 1.2.$$

(c) $\text{Var}(D) = 1.2$ is larger than 0.9. The defect rate is itself random, and this second source of randomness adds to the variance of D .

(d) By the law of total probability, $\mathbb{P}(D = 0) = \int_0^{1/5} \mathbb{P}(D = 0 \mid \Theta = \theta) f_\Theta(\theta) d\theta = \int_0^{1/5} 5(1 - \theta)^{10} d\theta = \frac{5}{11} (1 - 0.8^{11})$.

$\mathbb{P}(D = 0) \approx 0.416$, against $0.9^{10} \approx 0.349$: a random rate makes a defect-free sample more likely than a fixed rate of 0.1 does.

Remark: each value $\mathbb{P}(D = d)$ is another such integral, so computing $\mathbb{E}[D]$ from the pmf of D would need eleven integrals; the law of total expectation in (a) needed one.

2. Waiting for a pattern. Toss a fair coin repeatedly. Fix a pattern of three letters, for example THH. Let T be the number of the toss on which the pattern is first completed. Example: for the tosses H, T, H, H, the pattern THH is completed on toss 4, so $T = 4$. The aim is $\mathbb{E}[T]$ for each of the three patterns THH, THT and TTT.

The method is the one used for HH in the lecture (Application 2). At every moment the game is in one of three states, which record how much of the pattern the most recent tosses have completed. For THH and for THT the states are: state 0, no progress; state 1, the last toss is T; state 2, the last two tosses are TH. For TTT the states are: state 0, no progress; state 1, the last toss is T; state 2, the last two tosses are TT. Let m_j be the expected number of further tosses until the pattern is completed, starting from state j . Then $\mathbb{E}[T] = m_0$.

- (a) For the pattern THH, condition on the next toss to write one equation for each of m_0, m_1, m_2 . Solve the three equations.
- (b) Do the same for THT.
- (c) Do the same for TTT.
- (d) State which of the three patterns has the smallest $\mathbb{E}[T]$.
- (e) The three patterns have the same length. Explain in at most three sentences why their expected waiting times differ. Hint: when an attempt at the pattern fails, how much of the pattern is still in place?

Solution. (a) THH. From state 0, a T moves to state 1 and an H stays in state 0. From state 1, an H moves to state 2 and a T stays in state 1. From state 2, an H completes the pattern and a T moves back to state 1. Each toss costs 1, so $m_0 = 1 + \frac{1}{2}(m_0 + m_1)$, $m_1 = 1 + \frac{1}{2}(m_1 + m_2)$, $m_2 = 1 + \frac{1}{2}(0 + m_1)$. The third equation gives $m_2 = 1 + \frac{1}{2}m_1$. The second gives $m_1 = 2 + m_2 = 3 + \frac{1}{2}m_1$, so $m_1 = 6$ and $m_2 = 4$. The first gives $m_0 = 2 + m_1 = 8$.

$(m_0, m_1, m_2) = (8, 6, 4)$, so $\mathbb{E}[T] = 8$ for THH.

(b) THT. The equations for m_0 and m_1 are the same as in (a). From state 2, a T completes the pattern and an H gives THH, which contains no start of THT, so the game returns to state 0: $m_2 = 1 + \frac{1}{2}(0 + m_0)$. Then $m_1 = 2 + m_2 = 3 + \frac{1}{2}m_0$ and $m_0 = 2 + m_1 = 5 + \frac{1}{2}m_0$, so $m_0 = 10$, $m_1 = 8$, $m_2 = 6$.

$(m_0, m_1, m_2) = (10, 8, 6)$, so $\mathbb{E}[T] = 10$ for THT.

(c) TTT. From state 0, a T moves to state 1 and an H stays. From state 1, a T moves to state 2 and an H returns to state 0. From state 2, a T completes the pattern and an H returns to state 0: $m_0 = 1 + \frac{1}{2}(m_0 + m_1)$, $m_1 = 1 + \frac{1}{2}(m_0 + m_2)$, $m_2 = 1 + \frac{1}{2}(0 + m_0)$. The first gives $m_0 = 2 + m_1$. Substituting $m_2 = 1 + \frac{1}{2}m_0$ into the second, $m_1 = \frac{3}{2} + \frac{3}{4}m_0$. Then $m_0 = \frac{7}{2} + \frac{3}{4}m_0$, so $m_0 = 14$, $m_1 = 12$, $m_2 = 8$.

$(m_0, m_1, m_2) = (14, 12, 8)$, so $\mathbb{E}[T] = 14$ for TTT.

(d) THH has the smallest expected waiting time: 8 tosses, against 10 for THT and 14 for TTT.

(e) The patterns differ in what survives a failed attempt. For THH, the toss that spoils TH is a T, and that T is the first letter of a new attempt; for THT, the spoiling toss is an H, which is not the first letter, so the game returns to state 0; for TTT, an H at any stage returns the game to state 0. The less progress a failure destroys, the shorter the expected wait.

3. Sum and difference of two dice. Two fair dice are rolled. Y is the first die and Z the second. Let $X = Y + Z$ and $D = Y - Z$. The two-dice table has 36 cells (y, z) with $y, z \in \{1, \dots, 6\}$, each of probability $\frac{1}{36}$. Recall $\mathbb{E}[X] = 7$ and $\mathbb{E}[D] = 0$.
- (a) Fix $x \in \{2, \dots, 12\}$. The event $\{X = x\}$ is the set of cells (y, z) with $y + z = x$. Show that if the cell (y, z) belongs to $\{X = x\}$ then so does the cell (z, y) , and that D takes opposite values on these two cells. Deduce that $\mathbb{E}[D \mid X = x] = 0$ for every x , so that $\mathbb{E}[D \mid X] = 0$.
 - (b) Fix $d \in \{-5, \dots, 5\}$. The event $\{D = d\}$ is the set of cells (y, z) with $y - z = d$. List these cells and the value of X on each. Show that $\mathbb{E}[X \mid D = d] = 7$ for every d , so that $\mathbb{E}[X \mid D] = 7$.
 - (c) Use (b), “taking out what is known” and the tower property to show that $\mathbb{E}[XD] = 0$. Deduce that $\text{Cov}(X, D) = 0$.
 - (d) Show that X and D are not independent. Use either of the following two methods. Method 1: show that X and D always have the same parity, then compare $\mathbb{P}(D \text{ odd}, X \text{ even})$ with $\mathbb{P}(D \text{ odd})\mathbb{P}(X \text{ even})$. Method 2: compute $\mathbb{P}(D = 5, X = 7)$ and compare it with $\mathbb{P}(D = 5)\mathbb{P}(X = 7)$.
 - (e) The lecture states (Property 3, “What Independence Does Not Say”) that $\mathbb{E}[X \mid D] = \mathbb{E}[X]$ does not imply $X \perp\!\!\!\perp D$. Explain in at most three sentences how (b) and (d) together give an example of this.

Solution. (a) If $y + z = x$ then $z + y = x$, so (z, y) belongs to $\{X = x\}$ whenever (y, z) does. On (y, z) the value of D is $y - z$; on (z, y) it is $z - y = -(y - z)$. The cells of $\{X = x\}$ therefore pair off into cells with opposite values of D (a cell with $y = z$ pairs with itself and has $D = 0$), and all cells have the same probability. So the average of D over the row $\{X = x\}$ is 0.

$\mathbb{E}[D \mid X = x] = 0$ for every x , hence $\mathbb{E}[D \mid X] = 0$.

(b) The cells of $\{D = d\}$ are $(y, y - d)$ for those y with both y and $y - d$ in $\{1, \dots, 6\}$; on such a cell $X = y + (y - d) = 2y - d$. For $d \geq 0$, y runs over $d + 1, \dots, 6$; for $d < 0$, over $1, \dots, 6 + d$. In both cases the values of y form a block of consecutive integers whose midpoint is $\frac{(d+1)+6}{2} = \frac{7+d}{2}$ (for $d \geq 0$) or $\frac{1+(6+d)}{2} = \frac{7+d}{2}$ (for $d < 0$). The cells have equal probability, so the average of y over the row is $\frac{7+d}{2}$, and the average of $X = 2y - d$ is $2 \cdot \frac{7+d}{2} - d = 7$. Example, $d = 5$: the single cell $(6, 1)$, where $X = 7$.

$\mathbb{E}[X \mid D = d] = 7$ for every d , hence $\mathbb{E}[X \mid D] = 7$.

(c) By the tower property, $\mathbb{E}[XD] = \mathbb{E}[\mathbb{E}[XD \mid D]]$. Taking out what is known, $\mathbb{E}[XD \mid D] = D \mathbb{E}[X \mid D] = 7D$ by (b). So $\mathbb{E}[XD] = \mathbb{E}[7D] = 7\mathbb{E}[D] = 0$.

$\text{Cov}(X, D) = \mathbb{E}[XD] - \mathbb{E}[X]\mathbb{E}[D] = 0 - 7 \cdot 0 = 0$.

(d) Method 1: $X = D + 2Z$, so $X - D = 2Z$ is even and X, D have the same parity. Hence $\mathbb{P}(D \text{ odd}, X \text{ even}) = 0$, while $\mathbb{P}(D \text{ odd}) > 0$ and $\mathbb{P}(X \text{ even}) > 0$, so the product is positive and independence fails. Method 2: $\{D = 5, X = 7\}$ is the single cell $(6, 1)$, so $\mathbb{P}(D = 5, X = 7) = \frac{1}{36}$; but $\mathbb{P}(D = 5)\mathbb{P}(X = 7) = \frac{1}{36} \cdot \frac{6}{36} = \frac{1}{216}$.

X and D are not independent.

(e) By (b), every row $\{D = d\}$ has the same average $7 = \mathbb{E}[X]$, which is the statement $\mathbb{E}[X \mid D] = \mathbb{E}[X]$. Independence would require every row to have the same distribution as X , which is impossible for a row of fewer than 11 cells, and (d) confirms that the distributions differ. Equal row averages are a weaker condition than equal row distributions.

4. A random number of rolls. Roll a fair die repeatedly. Let $X_1, X_2, \dots$ be the rolls: independent, each uniform on $\{1, \dots, 6\}$, with $\mu = \mathbb{E}[X_i] = 3.5$. Let N be the number of the first

roll that shows 5 or 6. Example: the rolls $2, 4, 1, 6, \dots$ give $N = 4$. Since each roll shows 5 or 6 with probability $\frac{1}{3}$, N is geometric with success probability $\frac{1}{3}$, and $\mathbb{E}[N] = 3$. The value of N is determined by the rolls, so N is not independent of the X_i , and the random-sums theorem of the lecture does not apply. Define

$$\begin{aligned} S &= X_1 + \dots + X_N && \text{(the stopping roll included),} \\ S' &= X_1 + \dots + X_{N-1} && \text{(the stopping roll excluded; } S' = 0 \text{ if } N = 1). \end{aligned}$$

- Compute $\mathbb{E}[S]$ by conditioning on the first roll X_1 . Hint: on the event $\{X_1 = k\}$ with $k \leq 4$, the rolls $X_2, X_3, \dots$ are again a sequence of fair rolls, so $X_2 + \dots + X_N$ has the same distribution as S (restart property).
- Compute $\mu \mathbb{E}[N]$. State whether it equals $\mathbb{E}[S]$.
- Compute $\mathbb{E}[S']$ by the method of (a).
- Compute $\mu \mathbb{E}[N - 1]$. State whether it equals $\mathbb{E}[S']$.
- Compute $\mathbb{E}[X_N]$, the mean of the stopping roll. Then explain in at most three sentences why the product “ μ times the expected number of terms” gives the correct value in (b) but not in (d).

Solution. (a) Condition on $X_1 = k$, each value having probability $\frac{1}{6}$. If $k \in \{5, 6\}$, the game stops and $S = k$. If $k \leq 4$, then $S = k + S''$ where $S'' = X_2 + \dots + X_N$ has the distribution of S by the restart property, so $\mathbb{E}[S \mid X_1 = k] = k + \mathbb{E}[S]$. By total expectation, $\mathbb{E}[S] = \frac{1}{6}(5 + 6) + \frac{1}{6} \sum_{k=1}^4 (k + \mathbb{E}[S]) = \frac{11}{6} + \frac{10}{6} + \frac{4}{6}\mathbb{E}[S]$. Solving, $\frac{2}{6}\mathbb{E}[S] = \frac{21}{6}$.

$$\mathbb{E}[S] = \frac{21}{2} = 10.5.$$

(b) $\mu \mathbb{E}[N] = 3.5 \cdot 3 = 10.5$. It equals $\mathbb{E}[S]$.

(c) Condition on $X_1 = k$ again. If $k \in \{5, 6\}$ then $N = 1$ and $S' = 0$. If $k \leq 4$ then $S' = k + S'''$ with S''' distributed as S' , so $\mathbb{E}[S' \mid X_1 = k] = k + \mathbb{E}[S']$. Hence $\mathbb{E}[S'] = \frac{1}{6} \sum_{k=1}^4 (k + \mathbb{E}[S']) = \frac{10}{6} + \frac{4}{6}\mathbb{E}[S']$, so $\frac{2}{6}\mathbb{E}[S'] = \frac{10}{6}$.

$$\mathbb{E}[S'] = 5.$$

(d) $\mu \mathbb{E}[N - 1] = 3.5 \cdot 2 = 7$. It does not equal $\mathbb{E}[S'] = 5$.

(e) The stopping roll X_N is 5 or 6, each with probability $\frac{1}{2}$.

$$\mathbb{E}[X_N] = 5.5; \text{ consistently, } \mathbb{E}[S'] = \mathbb{E}[S] - \mathbb{E}[X_N] = 10.5 - 5.5 = 5.$$

In (b) the decision to stop after roll n depends only on rolls $1, \dots, n$, and for such stopping rules the formula $\mathbb{E}[S] = \mu \mathbb{E}[N]$ still holds (Wald's identity, mentioned in the lecture). In (d) the last roll is excluded because of its value, so the $N - 1$ terms of S' are not typical rolls but rolls that showed at most 4, with mean 2.5 rather than μ . The formula treats them as typical rolls, which is why it gives 7 instead of 5.

Part B: Entropy

Logarithms are base 2, the unit is the bit, and $0 \log 0 = 0$. Notation as in the Entropy lecture. p_X is the pmf of X . $H(X)$ is the entropy of X , and $H(p)$ the entropy of a pmf p ; $h(p) = p \log \frac{1}{p} + (1 - p) \log \frac{1}{1-p}$ is the binary entropy function. For a prefix-free code c with codeword lengths $\ell_c(x)$, $L(c) = \sum_x p_X(x) \ell_c(x)$ is its average length, and L^* is the smallest average length over all prefix-free codes. $H(P, Q) = \sum_x P(x) \log \frac{1}{Q(x)}$ is the cross-entropy of the truth P with the code or model Q , and $D(P \parallel Q) = H(P, Q) - H(P)$ is the relative entropy. $H(X \mid Y)$ is conditional entropy and $I(X; Y) = H(X) - H(X \mid Y)$ is mutual information.

5. Codes for four symbols. A prefix-free code for four symbols has a length profile $(\ell_1, \ell_2, \ell_3, \ell_4)$, the lengths of its four codewords, and its Kraft sum is $\sum_{i=1}^4 2^{-\ell_i}$. The rover's code 0, 10, 110, 111 has profile $(1, 2, 3, 3)$. The fixed-length code 00, 01, 10, 11 has profile $(2, 2, 2, 2)$.

- (a) Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$. Using Kraft's inequality and Gibbs' lemma, show that every prefix-free code for four symbols has $L(c) \geq 1.75$ under p . Deduce that the rover's code is optimal for p .
- (b) Show that the only length profiles for four symbols with Kraft sum exactly 1 are $(2, 2, 2, 2)$ and $(1, 2, 3, 3)$, up to the order of the lengths. Hint: order the lengths $\ell_1 \leq \ell_2 \leq \ell_3 \leq \ell_4$ and treat the cases $\ell_1 \geq 2$ and $\ell_1 = 1$ separately.
- (c) Take as given that an optimal code has Kraft sum exactly 1 (if the sum were smaller, some codeword could be shortened). Let p be any pmf on four symbols, with its probabilities written in decreasing order $p_{(1)} \geq p_{(2)} \geq p_{(3)} \geq p_{(4)}$. Show that

$$L^* = \min(2, p_{(1)} + 2p_{(2)} + 3p_{(3)} + 3p_{(4)}).$$

- (d) Apply (c) to $p = (0.45, 0.25, 0.15, 0.15)$. State which code is optimal and the value of L^* . Compute $H(p)$.
- (e) Apply (c) to $p = (0.3, 0.3, 0.2, 0.2)$. State which code is optimal and the value of L^* . Compute $H(p)$.
- (f) The source coding theorem states $H(p) \leq L^* < H(p) + 1$. For each of the two pmfs in (d) and (e), write down the interval for L^* that this gives. For each pmf, state whether the average lengths of both candidate codes (fixed-length and rover's) lie inside the interval.

Solution. (a) Put $a(x) = 2^{-\ell_c(x)}$ for each symbol x . Kraft's inequality gives $\sum_x a(x) \leq 1$, and $\ell_c(x) = \log \frac{1}{a(x)}$. Gibbs' lemma with weights p then gives $L(c) = \sum_x p(x) \log \frac{1}{a(x)} \geq \sum_x p(x) \log \frac{1}{p(x)} = H(p) = 1.75$, with equality if and only if $a = p$, that is, lengths $(1, 2, 3, 3)$. The rover's code has these lengths and $L = 1.75$.

The rover's code is optimal for p , with $L^* = 1.75$.

(b) Order the lengths $\ell_1 \leq \ell_2 \leq \ell_3 \leq \ell_4$. Case $\ell_1 \geq 2$: every term satisfies $2^{-\ell_i} \leq \frac{1}{4}$, so the Kraft sum is at most $4 \cdot \frac{1}{4} = 1$, with equality only when every length is 2. This gives $(2, 2, 2, 2)$. Case $\ell_1 = 1$: the first term is $\frac{1}{2}$, so the other three must sum to $\frac{1}{2}$. If $\ell_2 = 1$ the first two terms already sum to 1 and nothing is left for ℓ_3, ℓ_4 . If $\ell_2 \geq 3$ the last three terms sum to at most $3 \cdot \frac{1}{8} = \frac{3}{8} < \frac{1}{2}$. So $\ell_2 = 2$, and then $2^{-\ell_3} + 2^{-\ell_4} = \frac{1}{4}$ with $\ell_3, \ell_4 \geq 3$ (since $\ell_3 \geq \ell_2 = 2$ and $\ell_3 = 2$ would give $\frac{1}{4} + 2^{-\ell_4} > \frac{1}{4}$), which forces $\ell_3 = \ell_4 = 3$. This gives $(1, 2, 3, 3)$.

The only profiles with Kraft sum 1 are $(2, 2, 2, 2)$ and $(1, 2, 3, 3)$.

(c) By the given fact and (b), an optimal code has profile $(2, 2, 2, 2)$ or $(1, 2, 3, 3)$. Profile $(2, 2, 2, 2)$ gives $L = 2$ for every p . For profile $(1, 2, 3, 3)$, the average length is smallest when the shortest codewords go to the most probable symbols: if a more probable symbol had a longer codeword than a less probable one, swapping the two codewords would not increase L . So the best code with this profile has $L = p_{(1)} + 2p_{(2)} + 3p_{(3)} + 3p_{(4)}$. The optimal code is the better of the two.

$$L^* = \min(2, p_{(1)} + 2p_{(2)} + 3p_{(3)} + 3p_{(4)}).$$

(d) $p = (0.45, 0.25, 0.15, 0.15)$: the profile $(1, 2, 3, 3)$ gives $L = 0.45 + 0.5 + 0.45 + 0.45 = 1.85 < 2$.

The rover's code (assigning 0 to the symbol of probability 0.45, 10 to the symbol of probability 0.25) is optimal, $L^* = 1.85$, and $H(p) = 1.840$.

(e) $p = (0.3, 0.3, 0.2, 0.2)$: the profile $(1, 2, 3, 3)$ gives $L = 0.3 + 0.6 + 0.6 + 0.6 = 2.1 > 2$.

The fixed-length code is optimal, $L^* = 2$, and $H(p) = 1.971$.

(f) For (d) the theorem gives $L^* \in [1.840, 2.840]$; both 1.85 and 2 lie in this interval. For (e) it gives $L^* \in [1.971, 2.971]$; both 2.1 and 2 lie in this interval. In both cases the theorem does not separate the two candidates, and the enumeration in (b) was needed.

6. Two coins. Coin A shows heads with probability 0.9. Coin B is fair. One of the two coins is chosen uniformly at random; call the chosen coin $C \in \{A, B\}$. The chosen coin is tossed once; call the result $X \in \{H, T\}$.

- Compute the pmf of X and the entropy $H(X)$.
- Compute $H(X | C = A)$, $H(X | C = B)$ and $H(X | C)$.
- State which is larger, $H(X)$ or $H(X | C)$. The inequality between them is an instance of $h(\frac{1}{2} \cdot 0.9 + \frac{1}{2} \cdot 0.5) \geq \frac{1}{2}h(0.9) + \frac{1}{2}h(0.5)$. Name the property of h , listed on the lecture's slide "Binary Entropy Function", that this inequality expresses.
- Compute $I(X; C) = H(X) - H(X | C)$.
- Compute $\mathbb{P}(C = A | X = H)$ and $\mathbb{P}(C = A | X = T)$. Use them to compute $H(C | X)$. Check that $H(C) - H(C | X)$ equals the value of $I(X; C)$ found in (d).

Solution. (a) $\mathbb{P}(X = H) = \mathbb{P}(C = A) \cdot 0.9 + \mathbb{P}(C = B) \cdot 0.5 = \frac{1}{2} \cdot 0.9 + \frac{1}{2} \cdot 0.5 = 0.7$, so $p_X = (0.7, 0.3)$ on (H, T) .

$H(X) = h(0.7) = 0.881$ bits.

(b) Given $C = A$, X has pmf $(0.9, 0.1)$, so $H(X | C = A) = h(0.9) = 0.469$. Given $C = B$, X has pmf $(0.5, 0.5)$, so $H(X | C = B) = h(0.5) = 1$. Averaging over C with weights $\frac{1}{2}, \frac{1}{2}$: $H(X | C) = \frac{1}{2}(0.469 + 1) = 0.734$ bits.

(c) $H(X) = 0.881$ is larger than $H(X | C) = 0.734$. The inequality $h(\frac{1}{2} \cdot 0.9 + \frac{1}{2} \cdot 0.5) \geq \frac{1}{2}h(0.9) + \frac{1}{2}h(0.5)$ is the concavity of h .

(d) $I(X; C) = 0.881 - 0.734 = 0.147$ bits. This is the information that one toss carries about which coin was chosen.

(e) By Bayes' rule, $\mathbb{P}(C = A | X = H) = \frac{\frac{1}{2} \cdot 0.9}{0.7} = \frac{0.45}{0.7} = \frac{9}{14} = 0.643$ and $\mathbb{P}(C = A | X = T) = \frac{\frac{1}{2} \cdot 0.1}{0.3} = \frac{0.05}{0.3} = \frac{1}{6}$. Then $H(C | X) = \mathbb{P}(X = H)h(\frac{9}{14}) + \mathbb{P}(X = T)h(\frac{1}{6}) = 0.7 \cdot 0.940 + 0.3 \cdot 0.650 = 0.853$ bits.

Since C is uniform on two values, $H(C) = 1$, and $H(C) - H(C | X) = 1 - 0.853 = 0.147 = I(X; C)$, as required.

7. Two missions. The rover has two possible missions, $M \in \{1, 2\}$, each with probability $\frac{1}{2}$. On mission m the instructions $X \in \{\text{up}, \text{down}, \text{left}, \text{right}\}$ have pmf P_m , where

$$P_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}), \quad P_2 = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}) \quad \text{on } (\text{up}, \text{down}, \text{left}, \text{right}).$$

Note that $H(P_1) = H(P_2) = 1.75$ bits. One code, built for a pmf Q , must be installed before the mission is known. On mission m it costs $H(P_m, Q)$ bits per instruction, so its expected cost is $\frac{1}{2}H(P_1, Q) + \frac{1}{2}H(P_2, Q)$.

- Show that the expected cost equals $H(\bar{P}, Q)$, where $\bar{P} = \frac{1}{2}(P_1 + P_2)$. Compute $\bar{P}$.
- Find the pmf Q that minimises the expected cost. Compute the minimal expected cost.

- (c) Compute the expected cost for $Q = P_1$ (the old code, built for mission 1, is kept) and for $Q = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ (the fixed-length code). Compare both with the minimal cost from (b) and with $\frac{1}{2}H(P_1) + \frac{1}{2}H(P_2) = 1.75$, the cost if the mission were known before the code was chosen.
- (d) On mission m the code of (b) wastes $D(P_m \parallel \bar{P})$ bits per instruction. Show that the expected waste $\frac{1}{2}D(P_1 \parallel \bar{P}) + \frac{1}{2}D(P_2 \parallel \bar{P})$ equals $I(M; X)$, the mutual information between the mission M and one instruction X . Compute it.
- (e) Codeword lengths must be integers, so the code of (b) cannot be installed as it stands. Problem 5(b) showed that the only length profiles for four symbols with Kraft sum 1 are $(2, 2, 2, 2)$ and $(1, 2, 3, 3)$. Among codes that assign one codeword to each instruction, find the one with the smallest expected cost, and give that cost.
- (f) Name the method from the lecture (slide “Closing the Gap: Code Blocks of Symbols”) by which the cost per instruction can be brought closer to the minimal cost of (b).

Solution. (a) By the definition of cross-entropy, $\frac{1}{2} \sum_x P_1(x) \log \frac{1}{Q(x)} + \frac{1}{2} \sum_x P_2(x) \log \frac{1}{Q(x)} = \sum_x \frac{1}{2}(P_1(x) + P_2(x)) \log \frac{1}{Q(x)} = \sum_x \bar{P}(x) \log \frac{1}{Q(x)} = H(\bar{P}, Q)$.

Averaging the two pmfs coordinate by coordinate, $\bar{P} = (\frac{5}{16}, \frac{3}{16}, \frac{3}{16}, \frac{5}{16})$.

(b) By Gibbs' inequality, $H(\bar{P}, Q) \geq H(\bar{P})$, with equality if and only if $Q = \bar{P}$. So the minimising Q is $\bar{P}$, and the minimal cost is

$$H(\bar{P}) = 2 \cdot \frac{5}{16} \log \frac{16}{5} + 2 \cdot \frac{3}{16} \log \frac{16}{3} = 1.049 + 0.906 = 1.954 \text{ bits per instruction.}$$

(c) Old code, $Q = P_1$: the cost is $\frac{1}{2}H(P_1, P_1) + \frac{1}{2}H(P_2, P_1) = \frac{1}{2}(1.75 + 2.625) = 2.1875$ bits. Fixed-length code: every codeword has length 2, so the cost is 2 bits.

Ordering the four: known mission $1.75 <$ minimal cost $1.954 <$ fixed-length $2 <$ old code 2.1875 .

(d) Using $D(P \parallel Q) = H(P, Q) - H(P)$ and (a) with $Q = \bar{P}$, $\frac{1}{2}D(P_1 \parallel \bar{P}) + \frac{1}{2}D(P_2 \parallel \bar{P}) = H(\bar{P}, \bar{P}) - \frac{1}{2}H(P_1) - \frac{1}{2}H(P_2) = H(\bar{P}) - \frac{1}{2}(H(P_1) + H(P_2))$. With M uniform, the instruction X has pmf $\bar{P}$, so $H(X) = H(\bar{P})$; and given $M = m$ it has pmf P_m , so $H(X \mid M) = \frac{1}{2}H(P_1) + \frac{1}{2}H(P_2)$. The expected waste is therefore $H(X) - H(X \mid M) = I(M; X)$.

$$I(M; X) = 1.954 - 1.75 = 0.204 \text{ bits per instruction.}$$

(e) Profile $(2, 2, 2, 2)$: cost 2. Profile $(1, 2, 3, 3)$, with the codewords of lengths 1 and 2 on the two instructions of probability $\frac{5}{16}$: cost $\frac{5}{16} \cdot 1 + \frac{5}{16} \cdot 2 + \frac{3}{16} \cdot 3 + \frac{3}{16} \cdot 3 = \frac{33}{16} = 2.0625$.

The fixed-length code is the best single-instruction code, with cost 2 bits per instruction.

(f) Block coding: code blocks of several instructions at once, with a code built for the pmf $\bar{P}$ on blocks. By the block coding theorem, the cost per instruction then approaches 1.954.

8. A Markov text source. A binary text is modelled as a Markov chain $X_1, X_2, \dots$ on $S = \{0, 1\}$ with transition matrix

$$Q = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix},$$

where row x is the conditional pmf of X_{i+1} given $X_i = x$. The chain is started from its stationary pmf π , so that every X_i has pmf π . Recall from the lecture: the entropy rate is $H(X_{i+1} \mid X_i) = \sum_{x \in S} \pi(x) H(Q(x, \cdot))$, the monographic entropy is $H(\pi)$, and the perplexity of a model that needs b bits per symbol is 2^b .

- (a) Find the stationary pmf π .
- (b) Compute the entropy rate $H(X_{i+1} \mid X_i)$.

- (c) Compute the monographic entropy $H(\pi)$ and the difference $H(\pi) - H(X_{i+1} | X_i)$. State which quantity from the lecture this difference is.
- (d) Compute $H(X_1, X_2, X_3)$ by the chain rule, using the Markov property.
- (e) Compute $H(X_1, X_2, X_3)$ again by listing the probabilities of the eight possible strings $x_1 x_2 x_3$ and applying the definition of entropy.
- (f) Compute the perplexity of the Markov model, of the monographic model and of the uniform model.
- (g) A compressor built on the monographic model uses a codeword of length $\log \frac{1}{\pi(x)}$ for the symbol x at every position. Compute its cost in bits per symbol on this text. Compute its waste, that is, the cost minus the entropy rate.

Solution. (a) Stationarity balances the flow $0 \rightarrow 1$ against the flow $1 \rightarrow 0$: $\pi_0 \cdot 0.1 = \pi_1 \cdot 0.3$, so $\pi_0 = 3\pi_1$. With $\pi_0 + \pi_1 = 1$,

$$\pi = \left(\frac{3}{4}, \frac{1}{4}\right).$$

(b) Row 0 of Q has entropy $h(0.1) = 0.469$ and row 1 has entropy $h(0.3) = 0.881$. Weighting the rows by π ,

$$H(X_{i+1} | X_i) = \frac{3}{4} \cdot 0.469 + \frac{1}{4} \cdot 0.881 = 0.572 \text{ bits per symbol.}$$

(c) $H(\pi) = h(\frac{1}{4}) = 0.811$ bits. The difference is $0.811 - 0.572 = 0.239$ bits.

This difference is $I(X_i; X_{i+1})$, the mutual information of two consecutive symbols (slide “Entropy Rate at Most Monographic Entropy”).

(d) Chain rule: $H(X_1, X_2, X_3) = H(X_1) + H(X_2 | X_1) + H(X_3 | X_1, X_2)$. By the Markov property $H(X_3 | X_1, X_2) = H(X_3 | X_2)$, and by stationarity both conditional terms equal the entropy rate. So

$$H(X_1, X_2, X_3) = H(\pi) + 2 \cdot 0.572 = 0.811 + 1.144 = 1.955 \text{ bits.}$$

(e) The string abc has probability $\pi(a)Q(a,b)Q(b,c)$; for example, 000 has probability $0.75 \cdot 0.9 \cdot 0.9 = 0.6075$. For the strings $000, 001, 010, 011, 100, 101, 110, 111$, in this order, the eight probabilities are

$$0.6075, \quad 0.0675, \quad 0.0225, \quad 0.0525, \quad 0.0675, \quad 0.0075, \quad 0.0525, \quad 0.1225.$$

$$H(X_1, X_2, X_3) = \sum_{abc} \mathbb{P}(abc) \log \frac{1}{\mathbb{P}(abc)} = 1.955 \text{ bits, agreeing with (d).}$$

(f) Markov model: $2^{0.572} = 1.49$. Monographic model: $2^{0.811} = 1.75$. Uniform model: $2^1 = 2$.

(g) The symbol x occurs with frequency $\pi(x)$, so the cost is $\sum_x \pi(x) \log \frac{1}{\pi(x)} = H(\pi) = 0.811$ bits per symbol.

The waste is $0.811 - 0.572 = 0.239$ bits per symbol, the mutual information of (c).

Part C: Stochastic Processes

Notation as in the Stochastic Processes lecture. B is a standard Brownian motion, so B_t has the $N(0, t)$ distribution. $\mathbb{P}_i$ and $\mathbb{E}_i$ refer to a chain started at state i . In the gambler’s ruin, p is the probability of winning a round, $q = 1 - p$, and $r = q/p$. $\mathbf{P} = (p(i, j))$ is the transition matrix; row i is the conditional pmf of the next state given the current state i . π is a stationary distribution, $\pi\mathbf{P} = \pi$. For a target set A , $h_i = \mathbb{P}_i(\text{reach } A)$ is the hitting probability and $k_i = \mathbb{E}_i[\text{number of steps until } A \text{ is reached}]$ is the expected hitting time; both satisfy the first-step systems of the lecture. Φ is the cdf of $N(0, 1)$; the values needed below are

$$\Phi(0.100) = 0.540, \quad \Phi(0.500) = 0.691, \quad \Phi(0.674) = 0.750, \quad \Phi(1.000) = 0.841.$$

9. Hitting times of Brownian motion. The reflection principle states that for $a > 0$ and $t > 0$,

$$\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a).$$

For $a > 0$ let $T_a = \min\{s \geq 0 : B_s = a\}$ be the first time B reaches the level a . The event $\{T_a \leq t\}$ is the same as the event $\{\max_{s \leq t} B_s \geq a\}$.

- (a) Compute $\mathbb{P}(\max_{s \leq 4} B_s \geq 2)$.
- (b) Write $\mathbb{P}(T_1 \leq t)$ in terms of Φ . Evaluate it at $t = 1$, $t = 4$ and $t = 100$.
- (c) The median of T_1 is the value t with $\mathbb{P}(T_1 \leq t) = \frac{1}{2}$. Find it.
- (d) The scaling property of the lecture states that $(\frac{1}{a}B_{a^2t})_{t \geq 0}$ is again a Brownian motion. Use it to show that T_a has the same distribution as a^2T_1 . Give the median of T_a .
- (e) The lecture showed that $\mathbb{E}[T_1] = \infty$. Explain in at most three sentences how an infinite mean is consistent with the median found in (c). You may use the approximation $\mathbb{P}(T_1 > t) \approx \sqrt{2/(\pi t)}$ for large t , from the lecture.

Solution. (a) By the reflection principle, $\mathbb{P}(\max_{s \leq 4} B_s \geq 2) = 2\mathbb{P}(B_4 \geq 2)$. Since $B_4 \sim N(0, 4)$, $\mathbb{P}(B_4 \geq 2) = \mathbb{P}(N(0, 1) \geq 1) = 1 - \Phi(1)$. So the probability is $2(1 - \Phi(1))$.

$\mathbb{P}(\max_{s \leq 4} B_s \geq 2) = 0.317$ (the three-place table value $\Phi(1) = 0.841$ gives 0.318; the values below are exact to three places).

(b) $\mathbb{P}(T_1 \leq t) = \mathbb{P}(\max_{s \leq t} B_s \geq 1) = 2\mathbb{P}(B_t \geq 1)$. Since $B_t \sim N(0, t)$, $\mathbb{P}(B_t \geq 1) = 1 - \Phi(1/\sqrt{t})$. So $\mathbb{P}(T_1 \leq t) = 2(1 - \Phi(1/\sqrt{t}))$.

At $t = 1$: $2(1 - \Phi(1)) = 0.317$. At $t = 4$: $2(1 - \Phi(0.5)) = 0.617$. At $t = 100$: $2(1 - \Phi(0.1)) = 0.920$.

(c) $2(1 - \Phi(1/\sqrt{t})) = \frac{1}{2}$ holds if and only if $\Phi(1/\sqrt{t}) = \frac{3}{4}$, that is $1/\sqrt{t} = 0.674$, that is $t = 1/0.674^2$.

The median of T_1 is 2.20.

(d) Let $\tilde{B}_t = \frac{1}{a}B_{a^2t}$, a Brownian motion by the scaling property. $\tilde{B}$ reaches 1 at time t exactly when B reaches a at time a^2t . So the first time $\tilde{B}$ reaches 1 is T_a/a^2 , and this time has the distribution of T_1 because $\tilde{B}$ is a Brownian motion. Hence T_a has the distribution of a^2T_1 .

The median of T_a is $2.20a^2$.

(e) Half of the paths reach 1 by time 2.2, but the waiting time of the other half has a heavy tail: $\mathbb{P}(T_1 > t) = 2\Phi(1/\sqrt{t}) - 1 \approx \sqrt{2/(\pi t)}$ decays only like $t^{-1/2}$. The mean is $\mathbb{E}[T_1] = \int_0^\infty \mathbb{P}(T_1 > t) dt$, and $\int_0^\infty t^{-1/2} dt = \infty$. A median of 2.2 and an infinite mean are consistent because the rare long waits are long enough to make the integral diverge.

10. Bold play. A gambler wins each round with probability $p = 0.4$ and loses it with probability $q = 0.6$, so $r = q/p = 1.5$. The fortune starts at £3 and the target is £10; the game ends when the fortune reaches £0 or £10. Two strategies are compared. Timid play stakes £1 per round. From the lecture, with $N = 10$ and starting fortune i ,

$$h_i = \frac{r^i - 1}{r^N - 1}, \quad \mathbb{E}_i[T] = \frac{i}{q - p} - \frac{N}{q - p} \cdot \frac{1 - r^i}{1 - r^N},$$

where T is the number of rounds until the game ends. Bold play stakes, from fortune i , the amount $\min(i, 10 - i)$: as much as possible without overshooting the target. A win adds the stake to the fortune and a loss removes it. So the fortune moves from 3 to 6 (win) or 0 (loss); from 6 to 10 or 2; from 2 to 4 or 0; from 4 to 8 or 0; from 8 to 10 or 6.

- (a) Timid play: compute h_3 and $\mathbb{E}_3[T]$ from the two formulas above.
- (b) Bold play: draw the chain on the states $\{0, 2, 3, 4, 6, 8, 10\}$ with the transition probabilities. Write the first-step system for $h_i = \mathbb{P}_i(\text{reach } 10)$.
- (c) Solve the system of (b) for h_3 . Compare with the timid h_3 of (a).
- (d) Bold play: write the first-step system for $k_i = \mathbb{E}_i[\text{number of bets until the game ends}]$ and solve it for k_3 . Compare with the timid $\mathbb{E}_3[T]$ of (a).
- (e) Now let the game be fair, $p = q = \frac{1}{2}$. Solve the bold-play system of (b) for h_3 . For timid play the lecture gives $h_3 = \frac{3}{10}$. Explain in at most three sentences why the two values agree.

Solution. (a) With $r = 1.5$: $h_3 = \frac{1.5^3 - 1}{1.5^{10} - 1} = \frac{2.375}{56.665} = 0.042$. And $\mathbb{E}_3[T] = \frac{3}{0.2} - \frac{10}{0.2} \cdot \frac{1 - 1.5^3}{1 - 1.5^{10}} = 15 - 50 \cdot \frac{-2.375}{-56.665} = 15 - 2.10 = 12.9$.

Timid play from $\mathcal{L}3$: $h_3 = 0.042$ and $\mathbb{E}_3[T] = 12.9$ rounds.

(b) The chain: $3 \rightarrow 6$ with probability p , $3 \rightarrow 0$ with probability q ; $6 \rightarrow 10$ (p), $6 \rightarrow 2$ (q); $2 \rightarrow 4$ (p), $2 \rightarrow 0$ (q); $4 \rightarrow 8$ (p), $4 \rightarrow 0$ (q); $8 \rightarrow 10$ (p), $8 \rightarrow 6$ (q); 0 and 10 are absorbing. The first-step system is

$$h_0 = 0, h_{10} = 1, h_3 = ph_6, h_6 = p + qh_2, h_2 = ph_4, h_4 = ph_8, h_8 = p + qh_6.$$

(c) From $h_2 = ph_4 = p^2h_8$ we get $h_6 = p + qp^2h_8$, and then $h_8 = p + qh_6 = p + qp + q^2p^2h_8$. So $h_8 = \frac{p(1+q)}{1-q^2p^2} = \frac{0.4 \cdot 1.6}{1-0.0576} = \frac{0.64}{0.9424} = \frac{400}{589} = 0.679$. Then $h_6 = p + qp^2h_8 = \frac{274}{589} = 0.465$, $h_4 = ph_8 = \frac{160}{589} = 0.272$, $h_2 = ph_4 = \frac{64}{589} = 0.109$, and $h_3 = ph_6 = \frac{548}{2945}$.

Bold play from $\mathcal{L}3$: $h_3 = 0.186$, about four and a half times the timid 0.042.

(d) Each bet costs one step, so $k_0 = k_{10} = 0$, $k_3 = 1 + pk_6$, $k_6 = 1 + qk_2$, $k_2 = 1 + pk_4$, $k_4 = 1 + pk_8$, $k_8 = 1 + qk_6$. Substituting k_8 into k_4 , then into k_2 , then into k_6 : $k_6 = 1 + q + qp + qp^2 + q^2p^2k_6$, so $k_6 = \frac{1+q+qp+qp^2}{1-q^2p^2} = \frac{1.936}{0.9424} = \frac{1210}{589}$, and $k_3 = 1 + pk_6 = 1 + \frac{484}{589} = \frac{1073}{589}$.

Bold play from $\mathcal{L}3$: $k_3 = 1.82$ bets, against 12.9 rounds of timid play.

(e) With $p = q = \frac{1}{2}$ the formula of (c) gives $h_8 = \frac{\frac{1}{2} \cdot \frac{3}{2}}{1 - \frac{1}{16}} = \frac{4}{5}$, then $h_2 = \frac{1}{4}h_8 = \frac{1}{5}$, $h_6 = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{5} = \frac{3}{5}$ and $h_3 = \frac{1}{2}h_6 = \frac{3}{10}$.

Bold play in the fair game: $h_3 = \frac{3}{10}$, the same as timid play.

Every round of a fair game has expected gain 0 whatever the stake, so the expected fortune when the game ends equals the starting fortune: $\mathbb{E}[X_T] = 3$. The final fortune is 10 with probability h_3 and 0 otherwise, so $\mathbb{E}[X_T] = 10h_3$. Hence $h_3 = \frac{3}{10}$ for every strategy that stops only at 0 or 10.

11. PageRank. A random surfer on n pages moves as follows: with probability $d = 0.85$ it follows a uniformly chosen link out of the current page, and with probability $1 - d = 0.15$ it jumps to a uniformly chosen page. Its transition matrix is

$$\mathbf{P} = d\mathbf{H} + \frac{1-d}{n}\mathbf{J},$$

where $\mathbf{H}_{ij}$ is $1/(\text{number of links out of } i)$ if page i links to page j and 0 otherwise, and $\mathbf{J}$ is the matrix of ones. The PageRank of a page is its entry in the stationary distribution π of $\mathbf{P}$.

- (a) Three pages with links $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow A$. Write $\mathbf{H}$ and $\mathbf{P} = 0.85\mathbf{H} + 0.05\mathbf{J}$.
- (b) Solve $\pi\mathbf{P} = \pi$, $\pi_A + \pi_B + \pi_C = 1$, and rank the three pages.

- (c) Pages A and B each have a single in-link. Explain in at most two sentences why $\pi_A > \pi_B$.
- (d) Find the stationary distribution of $\mathbf{H}$ itself, that is, without damping ($d = 1$).
- (e) Add a fourth page D with the links $C \rightarrow D$ (so C now links to both A and D) and $D \rightarrow D$, and no other links out of D. Find the stationary distribution without damping ($d = 1$) and with $d = 0.85$ (now $n = 4$, so $\mathbf{P} = 0.85\mathbf{H} + 0.0375\mathbf{J}$).
- (f) Describe in at most three sentences what the page D does to the surfer when $d = 1$, and what the damping changes.

Solution. (a) A has two out-links, B and C one each:

$$\mathbf{H} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \text{ so } \mathbf{P} = \begin{pmatrix} 0.05 & 0.475 & 0.475 \\ 0.05 & 0.05 & 0.9 \\ 0.9 & 0.05 & 0.05 \end{pmatrix}.$$

(b) The columns of $\pi\mathbf{P} = \pi$ read $\pi_A = 0.05 + 0.85\pi_C$, $\pi_B = 0.05 + 0.425\pi_A$, $\pi_C = 0.05 + 0.425\pi_A + 0.85\pi_B$ (using $\pi_A + \pi_B + \pi_C = 1$ to collect the 0.05 terms). Substituting the first two into the third and solving, $\pi = (\frac{686}{1769}, \frac{380}{1769}, \frac{703}{1769}) = (0.388, 0.215, 0.397)$.

Ranking: C first, A second, B third.

(c) The in-link of A is the only link out of C, the top-ranked page, so A receives all of C's link mass. The in-link of B is one of two links out of A, so B receives only half of A's link mass.

(d) $\pi\mathbf{H} = \pi$ reads $\pi_A = \pi_C$, $\pi_B = \frac{1}{2}\pi_A$, $\pi_C = \frac{1}{2}\pi_A + \pi_B$, with $\pi_A + \pi_B + \pi_C = 1$.

Without damping, $\pi = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$.

(e) Without damping, D is an absorbing state: $\mathbf{H}_{DD} = 1$. The surfer returns to C repeatedly, and on each visit it moves to D with probability $\frac{1}{2}$, so it reaches D with probability 1 and then never leaves. The only stationary distribution is $\pi = (0, 0, 0, 1)$.

With $d = 0.85$, solving $\pi\mathbf{P} = \pi$ with $\sum \pi = 1$ gives $\pi = (0.101, 0.080, 0.148, 0.671)$ for (A, B, C, D).

(f) When $d = 1$, D is absorbing: the surfer eventually enters D and stays there, and every other page has rank 0. With damping, each visit to D lasts a geometric number of steps with mean $1/0.15 = 6.7$, and each random jump lands on D with probability $\frac{1}{4}$, so D still holds two thirds of the mass. Damping keeps every rank positive but does not remove the advantage of a self-linking page; search engines treat such pages separately.

12. Tennis at deuce. From deuce, the server wins each point with probability p and loses it with probability $q = 1 - p$, independently of the other points. The game is won by the first player to lead by two points. The states are D (deuce), I (advantage server), O (advantage receiver), W (game won by the server) and L (game lost by the server). From D the next state is I or O; from I it is W or D; from O it is D or L; W and L are absorbing.

(a) Write the transition matrix, with the states in the order D, I, O, W, L.

(b) Write the first-step system for $h_i = \mathbb{P}_i(\text{reach W})$. Solve it and show that $h_D = \frac{p^2}{p^2 + q^2}$.

(c) Write the first-step system for $k_i = \mathbb{E}_i[\text{number of points until the game ends}]$. Solve it and show that $k_D = \frac{2}{1 - 2pq}$.

(d) Evaluate h_D and k_D for $p = 0.5, 0.55, 0.6$ and 0.7 . For $p = 0.55$, state h_D as a percentage.

- (e) Let $m \geq 1$. The game ends after exactly $2m$ further points if the first $m - 1$ pairs of points return to deuce and the m -th pair does not. Write h_D as an infinite series by summing over m . Then explain in at most two sentences why the first-step system of (b) gives h_D without summing a series.

Solution. (a) In the order D, I, O, W, L:

$$\mathbf{P} = \begin{pmatrix} 0 & p & q & 0 & 0 \\ q & 0 & 0 & p & 0 \\ p & 0 & 0 & 0 & q \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

(b) $h_W = 1$, $h_L = 0$, $h_D = ph_I + qh_O$, $h_I = p + qh_D$, $h_O = ph_D$. Substituting the last two equations into the third: $h_D = p(p + qh_D) + qph_D = p^2 + 2pq h_D$, so $h_D(1 - 2pq) = p^2$. Since $p + q = 1$, $1 - 2pq = (p + q)^2 - 2pq = p^2 + q^2$.

$$h_D = \frac{p^2}{p^2 + q^2}. \text{ For } p = 0.6: h_D = \frac{9}{13}, h_I = \frac{57}{65}, h_O = \frac{27}{65}.$$

(c) $k_W = k_L = 0$, $k_D = 1 + pk_I + qk_O$, $k_I = 1 + qk_D$, $k_O = 1 + pk_D$. Substituting: $k_D = 1 + p(1 + qk_D) + q(1 + pk_D) = 1 + p + q + 2pq k_D = 2 + 2pq k_D$.

$$k_D = \frac{2}{1 - 2pq}. \text{ For } p = 0.6: k_D = \frac{50}{13} = 3.85 \text{ points.}$$

(d) For $p = 0.5, 0.55, 0.6, 0.7$: $h_D = 0.5, 0.599, 0.692, 0.845$ and $k_D = 4, 3.96, 3.85, 3.45$.

For $p = 0.55$, the server wins the game from deuce with probability 60%: a 55/45 edge per point becomes a 60/40 edge per game, because the game ends only when one player is two points ahead.

(e) A pair of points returns to deuce with probability $2pq$ (one point each, in either order) and ends the game with probability $p^2 + q^2$, of which p^2 is a win for the server. So the server wins after exactly $2m$ points with probability $(2pq)^{m-1}p^2$, and

$$h_D = \sum_{m \geq 1} (2pq)^{m-1} p^2 = \frac{p^2}{1 - 2pq}, \text{ the same value as in (b).}$$

The first-step system uses the return to deuce to write h_D in terms of itself, in the equation $h_D = p^2 + 2pq h_D$. Solving this one equation performs the summation of the geometric series implicitly.